Understanding Transferable Urban Mobility Structure: Distance Decay Recovery and Interpretable Zero-Shot Prediction

Thinh Nguyen[1*]

[1*]Aarista Technologies, Ho Chi Minh City, Vietnam.

Corresponding author(s). E-mail(s): nqt900@gmail.com;

Abstract

Designing transferable mobility models for urban environments where no OD (origin-destination) records are available poses an unsolved problem. In order to tackle the problem of high costs and privacy concerns linked with mobility trace collection, recent works focus on leveraging open data sources to reduce the proprietary data. However, most of the approaches still use individual data, so they reduce privacy. Focusing on this issue, we introduce the Separable Decomposed Framework for Zero-Shot Transferable Flows (SDF-ZTF), which is a decomposed modeling framework for OD flow generation through integration of open-data-based features with a production-constrained gravity model. Using only globally accessible open datasets, including OpenStreetMap, WorldPop, and Meta AI for Good, our methodology allows generating estimates of mobility patterns for unseen cities without using any target-city-specific OD data or training. Experimental evaluation shows that our decomposition method matches current deep learning architectures in a zero-shot transfer setting and provides policy relevance.

Keywords: Urban mobility, origin-destination flows, gravity models, interpretability, transfer learning, zero-shot transfer, open-data modeling

1 Introduction

Prediction of origin-destination (OD) flows is an essential task for transportation planning and urban analysis (Barbosa et al. 2018; González et al. 2008; Song et al. 2010). Unfortunately, the traditional approach to collecting OD flows data is through conducting household travel surveys that are costly and time-consuming.

Two major types of OD flow prediction models were designed in order to overcome those drawbacks. The classical gravity and radiation models (Wilson 1971; Simini et al. 2012; Stouffer 1940) provide physically justifiable and interpretable prediction but need local data for calibration. Deep learning OD flow prediction models such as DeepGravity (Simini et al. 2021) and graph neural network-based models (Rong et al. 2023; Wang et al. 2021), on the other hand, provide superior prediction performance but remain black box and still need the target city data.

Data on aggregate trip length distribution, which is collected from mobile devices and location sharing applications, become available even when OD Matrix is not available. This kind of data does not include information about individual trips and can be used under strict anonymous

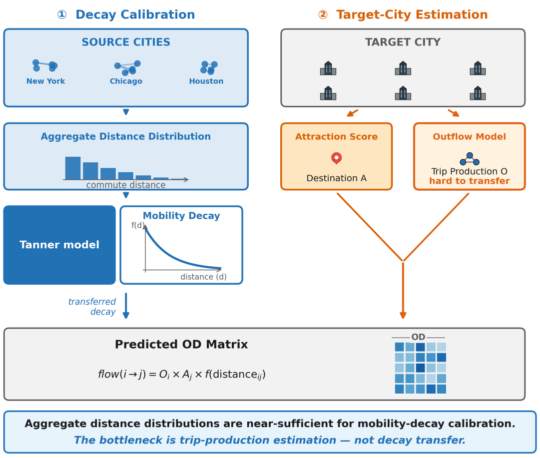

Fig. 1 SDF-ZTF framework: Three independent components (origin production O_i from features, distance decay $f(d)$ from aggregate data, destination attraction A_j from features)

policy. One can raise a question whether it is possible to calibrate the distance decay function using aggregate data in a new city and what part of spatial interaction function is more important for accurate prediction.

There are three research questions that will be explored in the paper:

1. Is it possible to use aggregate travel distance distribution data to calibrate the distance decay function without using individual OD matrix?
2. Which part of spatial interaction plays the key role in the process of accurate prediction: production, attraction or distance decay?
3. Is it possible to perform decomposition of interpretable model and reach neural baseline performance under zero-shot cross-city transfer using only open data?

In this work, we present the Separable Decomposed Framework for Zero-Shot Transferable Flows (SDF-ZTF), where OD flows are decomposed into three separate parts: origin production, distance decay, and destination attractiveness. SDF-ZTF makes use of solely global data sets—OpenStreetMap (Atwal et al. 2025) and WorldPop (Tatem 2017)—that can be deployed anywhere without conducting any kind of local survey and training. See Fig. 2 for comparison of the proposed zero-shot method with the common locally dependent approach.

Contributions of this paper include the following:

Decay Recovery: Distance distributions suffice to recover decay parameters without trajectories.
Ranking of Components: Distance decay is the component that explains 75.7% of explainable variance in cross-city transfer.
Zero-shot Equivalence: The decomposed framework performs similarly to DeepGravity in zero-shot settings but is interpretable and uses open data only.

2 Related Work

2.1 Classical and Structured Mobility Models

Gravity models (Wilson 1971; Tanner 1961) and radiation models (Simini et al. 2012; Alis et al. 2021) provide the basis of spatial interaction modeling. Gravity models employ physical metaphors to link flows to mass and distance in zones, while radiation models (Stouffer 1940) interpret the choice of destinations probabilistically on the basis of opportunities in between places. Scaling-law studies (Brockmann et al. 2006; González et al. 2008) and predictability analysis (Song et al. 2010) have shown that travel distances obey clear statistical patterns in urban settings. Interpretability and parsimony characterize both models, but gravity models need local OD information for decay parameters estimation, while radiation models work poorly in cities where commuters regularly avoid opportunities that are closer.

2.2 Deep Learning for OD Prediction

DeepGravity (Simini et al. 2021) uses multilayer perceptrons to associate urban characteristics with flow volumes. Recent advancements have included graph neural networks (Rong et al. 2023), spatio-temporal networks (Wang et al. 2021), and OSM-based commuter modeling (Atwal et al. 2025). These methods are capable of modeling non-linear associations and work well where training data is abundant. The problem is that these are black box models and perform poorly out of distribution, needing lots of OD data from target cities.

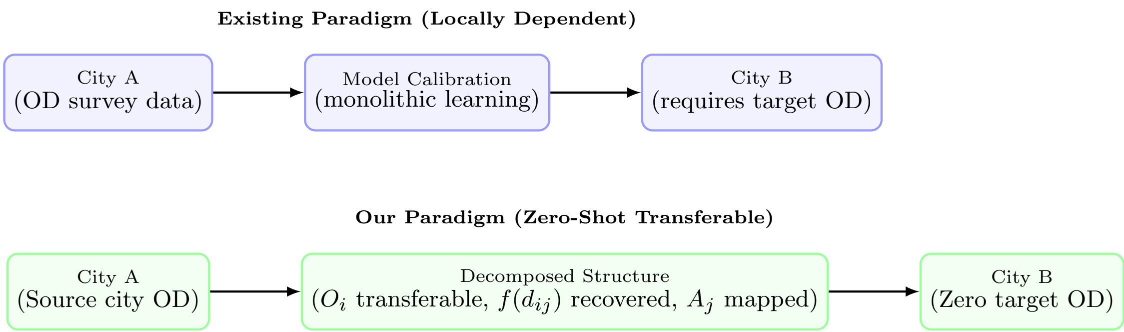

Fig. 2 Conceptual comparison of the existing locally dependent modeling paradigm versus our zero-shot transferable paradigm. Under the existing paradigm, models require target-city OD observations to retrain or calibrate parameters. In our paradigm, we learn transferable origin production structures from source City A, recover decay from target aggregate data, and map attraction from local open features in target City B, enabling zero-shot transfer without target-city OD surveys.

2.3 Aggregate Mobility Data for Decay Calibration

Today's location-aware mobile phones yield mobility signals of entire populations (Barbosa et al. 2018; González et al. 2008) that can be summarized to form trip-length distributions—histograms of traveled distances—that do not reveal any OD pairs (Pappalardo et al. 2023). Previous studies have proven that such distributions are indicative of the friction of urban trips and match distance-decay models (Brockmann et al. 2006; Lenormand et al. 2012, 2016). However, to the best of our knowledge, there is no existing study on using such distributions to calibrate zero-shot cross-city transfer.

2.4 Cross-City Transfer Learning

Transfer learning across cities is designed to forecast mobility in cities that lack OD survey data. The existing literature on cross-city transfer learning can be divided into three categories:

1. **Similarity-based**: RegionTrans (Wang et al. 2018) uses spatial similarity to align source and target regions; CrossTReS (Jin et al. 2022) re-weights source regions using meta-learning.
2. **Representation learning**: Geospatial embeddings and GeoAI foundation models (Mai et al. 2022; Mendieta et al. 2023) use representation learning to develop features for urban systems.
3. **Domain adaptation**: STAN (Wang et al. 2022) and meta-spatiotemporal networks (Yao et al. 2019) align trained models to the target domains.

All three types still need to have target-city observations, calibration, or fine-tuning.

2.5 Explainable and Transferable Urban Representations

SHAP (Lundberg and Lee 2017) is extensively employed for feature importance interpretation in spatial models, but nearly always restricted to just one city. If SHAP-based feature rankings are consistent across cities and therefore may help with feature selection, this question has not yet been systematically addressed.

2.6 Research Gap and Positioning

Table 1 compares existing model classes with our proposed SDF-ZTF.

Table 1 lists the current model classes and contrasts them against our proposed SDF-ZTF. The current literature fails to provide any approach that offers interpretability, open-data-only approach, zero-shot city transfer capability, and identification of transferable spatial features systematically. There are four gaps that this research fills:

G1: Calibration. The gravity model needs local OD survey data for decay parameter estimation.
G2: Interpretability. The neural models are not interpretable but accurate.
G3: Data-dependent. Even the transfer learning approaches need OD data of the target city.

Table 1 Comparative analysis of human mobility models across structural and transferability dimensions. "Calibration-Free Decay" indicates whether the model recovers the distance decay function without requiring individual OD pair observations from target cities.

Model Class	**Decomposed Structure**	**Cross-City Transfer**	**Zero-Shot Target**	**Interpretability**	**Calibration-Free Decay**
Gravity (Wilson 1971)	✓	×	×	✓	×
Radiation (Simini et al. 2012)	×	✓	✓	✓	✓
DeepGravity (Simini et al. 2021)	×	Limited	×	×	×
GNN OD (Rong et al. 2023)	×	Limited	×	×	×
RegionTrans (Wang et al. 2018)	Partial	✓	×	×	×
SDF-ZTF (Ours)	✓	✓	✓	✓	✓

G4: Feature Generalizability. Currently, there is no approach that identifies transferable spatial features systematically.

3 Methodological Design

3.1 Decomposition Hypothesis

We hypothesize that OD flows can be decomposed into three separable, city-independent components:

$$T_{ij} = O_i \cdot f(d_{ij}) \cdot A_j \tag{1}$$

where O_i represents origin production capacity, $f(d_{ij})$ represents distance decay, and A_j captures destination attraction. We hypothesize that: (1) each component is approximately independent, (2) the underlying mechanisms can be learned from spatial descriptors of the built environment, and (3) the learned origin production relationship transfers directly to unseen target cities, while decay parameters are recovered from target-city aggregate trip-length distributions, and attraction is computed from locally available open data. This decomposition hypothesis is directly tested by evaluating which of these three components exerts the dominant influence on predictive accuracy. Section 5 evaluates and validates this hypothesis empirically.

3.2 Experimental Framework for Investigating Transferable Mobility Structure

The conceptual pipeline of the proposed Separable Decomposed Framework for Zero-Shot Transferable Flows (SDF-ZTF) is illustrated in Fig. 3.

3.3 Distance Decay Function

We use the Tanner multiplicative deterrence function (Tanner 1961), which combines power-law and exponential terms:

$$f(d_{ij};\, \beta, \gamma, \delta) = (d_{ij} + \delta)^{-\gamma} \cdot \exp(-\beta\, d_{ij}) \tag{2}$$

where $\beta > 0$ is the exponential decay rate governing long-range trip attenuation, $\gamma \geq 0$ is the power-law exponent modeling short-to-medium range trip distribution, and $\delta > 0$ is the adaptive geometric anchor.

The offset δ prevents singularity at $d_{ij} = 0$ and adapts to zone size:

$$\delta = 0.5 \times \bar{R}_{\text{zone}}, \qquad \bar{R}_{\text{zone}} = \frac{1}{N}\sum_{i=1}^{N}\sqrt{\frac{\text{Area}_i}{\pi}} \tag{3}$$

where $\bar{R}_{\text{zone}}$ is the mean zone radius. Parameters (β, γ) are estimated via MLE on the aggregate distance-bin distribution. Calibration and optimization details are in the Appendix. Because only binned trip-length histograms are used—not individual OD records—this procedure is compatible with strict data-governance regulations (e.g., GDPR).

3.4 Origin Outflow Component (O_i)

The trip production $O_i = \sum_j T_{ij}$ is the total trips originating from zone i. We predict O_i with a GBDT regressor trained on 22 features selected via cross-city SHAP stability (Section 4) from a 30-dimensional candidate pool of open spatial data. Hyperparameters are in Appendix B.

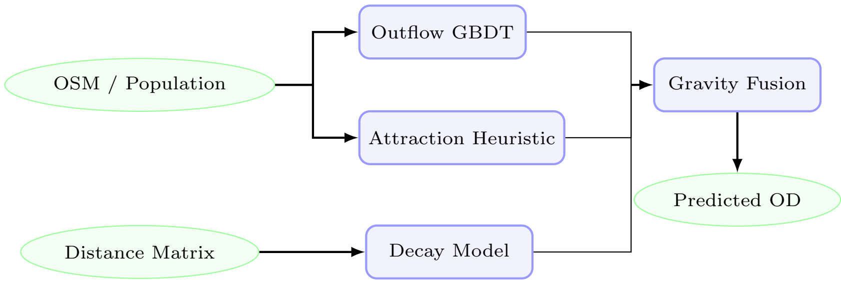

Fig. 3 Pipeline of the SDF-ZTF framework. Globally available spatial features are used to predict trip production (O_i) via a GBDT model and to compute destination attraction (A_j) exclusively via a training-free min-max heuristic (a GBDT attraction model is trained only during ablation validation to prove that this simple heuristic is sufficient, avoiding unnecessary complexity for cross-city transfer). Distance decay is estimated separately from aggregate distributions. The components are combined via multiplicative gravity fusion to predict zero-shot flow matrices (T_{ij}).

3.5 Destination Attraction Component (A_j)

Destination attraction A_j measures how attractive zone j is to travelers. We compute it with a training-free heuristic: $\hat{A}_j \propto \frac{1}{2}(\text{minmax}(P_j) + \text{minmax}(\text{POI}_j))$. Section 5 validates this choice by showing that a trained GBDT attraction model performs comparably, confirming that the heuristic is sufficient for cross-city transfer.

3.6 Algorithmic Flow

The complete pipeline for training and zero-shot deployment is formalized in Algorithm 1.

4 Data and Transferable Urban Representation

4.1 Design Principle: Open-Data-Only

All features in SDF-ZTF come from globally available open data rather than proprietary cellular records or household surveys. This makes the framework deployable in any city. We intentionally exclude local employment databases (e.g., LODES), which lack global equivalents, and instead proxy employment activity through commercial and industrial POI densities from OpenStreetMap.

4.2 Data Sources

We use three data sources. Their roles across framework components are shown in Table 2.

4.2.1 (1) Origin-Destination (OD) Matrix - T_{ij}

Seasonal OD flows between census tracts are extracted from Meta's Movement Distribution Maps (Meta ????) for 50 US metropolitan areas. The OD matrix serves only as training labels for source cities and as evaluation ground truth; it is never used as a predictive feature.

4.2.2 (2) OSM-Derived Features

We extract POIs and road features using the Overpass API from OpenStreetMap snapshots (Atwal et al. 2025). POI counts are calculated across 16 land-use categories (e.g., retail, industrial, education, health).

4.2.3 (3) Population Data

Population counts come from the original mobility dataset but can be replaced with WorldPop (Tatem 2017) gridded estimates (100 m resolution) in cities lacking census data.

4.3 Transferable Urban Features

We use a two-stage procedure—feature engineering followed by stability selection—to identify spatial patterns that generalize across cities. The feature taxonomy is shown in Fig. 4.

4.3.1 Candidate Feature Set (30 dimensions)

We initially construct 30 candidate features per zone. This candidate space comprises 23 scale and density variables (zone area, road length, residential population, total POI counts, and 16 POI

Algorithm 1: SDF-ZTF Training and Zero-Shot Prediction

Input: Source cities data $\{\mathbf{X}^{(c)}, \mathbf{T}^{(c)}\}_{c=1}^{C}$; Target city spatial features $\mathbf{X}^{(t)}$

Output: Predicted Target City OD Matrix $\hat{\mathbf{T}}^{(t)}$

1. **Decay Calibration**: Recover target-city decay parameters $(\hat{\beta}, \hat{\gamma})$ via MLE on the target city's aggregate distance-bin histogram (the type of population-level summary routinely available from mobile network operators; see Appendix A). In the absence of target-city aggregate data, a national prior $(\beta_{\text{nat}}, \gamma_{\text{nat}})$ derived from the pooled source-city histograms may be used as a fallback.
2. **Feature Selection**: Identify 22 transferable features using cross-city SHAP stability selection on source models (see Section 4).
3. **Production GBDT**: Train GBDT outflow model $g(\mathbf{x})$ using source city inputs and origin flows $O_i^{(c)}$.
4. **Attraction Heuristic**: Define the training-free attraction function $a(\mathbf{x}_j) = \frac{1}{2}\left(\text{minmax}(P_j) + \text{minmax}(\text{POI}_j)\right)$.
5. **Zero-Shot Target Prediction**:
   a. Predict target city outflows: $\hat{O}_i^{(t)} = g(\mathbf{x}_i^{(t)})$.
   b. Compute target city attractions: $\hat{A}_j^{(t)} = a(\mathbf{x}_j^{(t)})$.
   c. Compute adaptive target offset: $\delta^{(t)} = 0.5 \times \bar{R}_{\text{zone}}^{(t)}$.
   d. Fuse components: $\hat{T}_{ij}^{(t)} = \hat{O}_i^{(t)}(d_{ij} + \delta^{(t)})^{-\hat{\gamma}} e^{-\hat{\beta} d_{ij}} \hat{A}_j^{(t)}$.

Table 2 Data usage matrix illustrating the operational roles of inputs across SDF-ZTF components to guarantee leakage-free prediction.

Data Source	**Outflow** O_i	**Attraction** A_j	**Distance Decay** $f(d)$	**Evaluation**
OD Matrix (T_{ij})	×	×	✓	✓
Aggregate Trip-Length Distribution	×	×	✓	×
Distance Matrix (d_{ij})	×	×	✓	✓
OpenStreetMap (OSM)	✓	✓	×	×
WorldPop	✓	✓	×	×

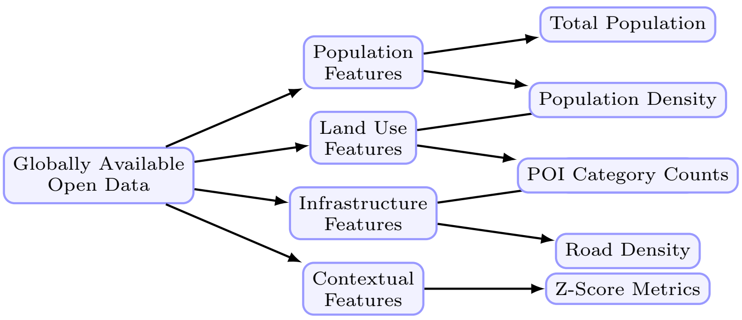

Fig. 4 Taxonomy of the globally available open data features used in the SDF-ZTF framework. Features are categorized into four dimensions: population, land use, infrastructure, and contextual scale factors.

categories) and 7 city-wide Z-score context features. The Z-score context features are defined as:

$$z_{ip} = \frac{x_{ip} - \mu_p^{(c)}}{\sigma_p^{(c)}} \tag{4}$$

where $\mu_p^{(c)}$ and $\sigma_p^{(c)}$ represent the mean and standard deviation of feature p across all zones in city c. These context features normalize absolute scale differences, capturing the relative hierarchy of zones within their respective metropolitan areas.

4.3.2 SHAP Stability Selection

Features are ranked by a cross-city stability score $S_p = \mu_p \,/\, (\sigma_p + \epsilon)$, where μ_p and σ_p denote the mean and standard deviation of per-city GBDT

SHAP importances and $\epsilon = 10^{-5}$. A feature is retained if $\mu_p \geq 0.005$ and $S_p \geq 1.0$, ensuring both relevance and cross-city consistency. This procedure selects 22 features (26.7% dimensionality reduction) without loss of zero-shot prediction accuracy (Appendix C).

4.4 Cross-City Transfer Evaluation Protocol

Standard mobility evaluations split OD pairs within a single city, which leaks spatial information from test zones into training. We instead use a strict cross-city protocol:

- **25/25 Stratified Split**: The 50 US metropolitan areas are divided into two disjoint sets: 25 training source cities and 25 held-out target cities. Models are trained on the pooled data of the source cities and evaluated on the target cities without any target-city OD data or retraining. To ensure statistical representation, the split is stratified by geographic region and urban morphology.
- **50-Fold Leave-One-City-Out (LOCO)**: The framework is iteratively trained on $C-1$ cities (49 cities) and evaluated on the single remaining held-out city, repeating this cycle for all 50 metropolitan areas to ensure robust out-of-distribution evaluation.

4.4.1 Information Leakage Prevention

Under this cross-city evaluation protocol, we enforce a strict information firewall between the source and target environments.

- **Forbidden**: Target-city flow matrices T_{ij}, target-city household survey results, target cellular data, and target trip-demand signals.
- **Allowed**: Centroid coordinates, administrative zone boundaries, OpenStreetMap POIs, and gridded open demography (WorldPop).

Since destination attraction is computed via a training-free open-data heuristic and distance decay parameters are learned on source cities, zero-shot leakage is mathematically prevented.

4.4.2 Evaluation Metric: Common Part of Commuters

We measure prediction quality with the Common Part of Commuters (CPC) (Lenormand et al. 2012):

$$\text{CPC}(T, \hat{T}) = \frac{2\sum_{i,j}\min(T_{ij}, \hat{T}_{ij})}{\sum_{i,j} T_{ij} + \sum_{i,j}\hat{T}_{ij}} \quad (5)$$

where T_{ij} represents the ground-truth commuter flow from zone i to zone j, and $\hat{T}_{ij}$ is the predicted flow. The CPC index ranges from 0 (indicating completely disjoint flow matrices) to 1 (indicating perfect prediction).

4.5 Preprocessing & Normalization

Data preprocessing details are summarized in Table 3.

5 Results

5.1 Framework for Validation Overview

SDF-ZTF is validated through three phases (Fig. 5): (1) validating each component separately, (2) calculating the individual contribution of each component by means of factorial ablation, and (3) comparing performance to the baselines. All models are trained on 25 source cities and evaluated on 25 held-out targets, with no target-city OD data.

5.2 Component Validation

The first phase includes the validation of each component separately.

5.2.1 Decay Recovery Validation

In this phase we compare two estimation modes for (γ, β) for the gravity model with production constraints, where A_j and O_i remain unchanged (OSM heuristic):

- Ground Truth (GT): Estimation is done using individual origin-destination (OD) pairs.
- Recovered: Estimation using aggregated distance bins distribution (trip length marginal histogram, 50 equal-width bins).

Further details can be found in Appendix A.4.

Table 3 Preprocessing and normalization steps applied to spatial features prior to model learning.

Step	Treatment	Rationale
Missing Data	Median imputation	Rare (< 2%); resolved from spatial neighbors
Log Transform	Population, POI counts	Stabilize right-skewed distributions
Standardization	Per-city (not global)	Preserve relative spatial hierarchy
Outliers	Retain	Important spatial hubs (CBDs); robust models

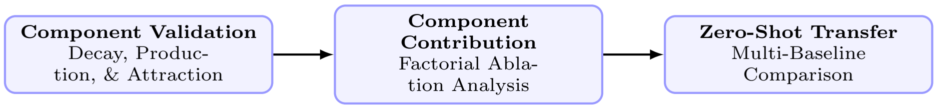

Fig. 5 Methodology validation roadmap showing the three evaluation stages used to systematically verify the SDF-ZTF framework.

Pure Decay Recovery We estimate the Tanner Multi decay function (see Eq. 2) on the aggregate distance distributions of the training data for each city. We compare the estimated parameters $(\hat{\gamma}_{\text{recovered}}, \hat{\beta}_{\text{recovered}})$ to the ground truth parameters $(\hat{\gamma}_{\text{gt}}, \hat{\beta}_{\text{gt}})$ estimated directly on OD pairs.

Result: The parameter recovery performance metrics are summarized in Table 4 across the 25 target cities.

Crucially, evaluating downstream OD prediction under oracle outflow (isolating the impact of the decay function) demonstrates that using the recovered parameters yields virtually identical performance to the ground truth parameters (Table 5). The mean CPC is 0.7411 under ground truth parameters and 0.7398 under recovered parameters, resulting in an extremely small average gap of just 0.0013 (0.18%). This confirms that aggregate-spaced recovery is functionally equivalent to fitting on individual OD trajectories for prediction.

The parameter recovery is strong for both γ ($r = 0.893$) and β ($r = 0.582$). A weaker recovery for β relative to γ is anticipated due to the noisy tail under aggregation binning. However, the downstream prediction evaluation proves that the recovered decay curve is robust and sufficient for prediction.

Bin Sensitivity Analysis (varying $K \in \{3, 5, 10, 20\}$ distance bins) is explained in Appendix A.2 and demonstrates that both parameters and downstream prediction accuracy stabilize for $K \geq 10$.

Additional robustness experiments (including sensitivity to distance bin configurations, missing attractiveness data, and CBD collapse scenarios) and the corresponding recovery error heatmap are detailed in Appendix A.2.

5.2.2 Origin Production Transferability

To control for the effect of O_i estimation, we hold A_j (OSM heuristic) and (γ, β) (national prior) fixed, then test the performance of three different outflow strategies in 25 held-out cities: a simple population baseline ($O_i \propto \text{Population}_i$), GBDT-predicted outflows, and oracle outflows. The results can be seen in Table 6.

GBDT outflows (CPC 0.7037) beat the simple baseline (0.584) and achieve 94.5% of the oracle upper bound (0.7447). Robustness experiments are shown in Appendix B.

5.2.3 Transferable Feature Discovery

Feature importance and feature stability is measured only with the 25 source cities (no target city leakage). Table 7 shows the list of top 10 features.

Population and activity-related points of interest are the most stable transferable predictors.

SHAP Feature Count Sweep. CPC plateaus at $K = 15$ features (0.7124) until $K = 22$. We use $K = 22$ as our number of features because it contains all features that pass both cutoffs ($\mu_p \geq 0.005$, $S_p \geq 1.0$). See Fig. 7.

5.3 Component Contributions Analysis

A factorial ablation over the 25 held out cities, with replacement of each disabled component with uniform weighting.

Table 4 Decay parameter recovery performance metrics and downstream CPC under oracle outflow across the 25 target cities.

Parameter	MAE	RMSE	Pearson r	Pearson r^2
Power-law exponent (γ)	0.0394	0.0549	0.893	0.798
Exponential decay rate (β, km^{-1})	0.0024	0.0052	0.582	0.338

Table 5 Downstream travel flow prediction performance (CPC) comparison between Ground Truth and Recovered decay calibration strategies.

Calibration Strategy	Data Requirement	Mean CPC	CPC Gap
Ground Truth (GT)	Individual OD Trajectories	0.7411	Reference
Recovered (Proposed)	Aggregate Distance Bins	0.7398	-0.0013 (-0.18%)

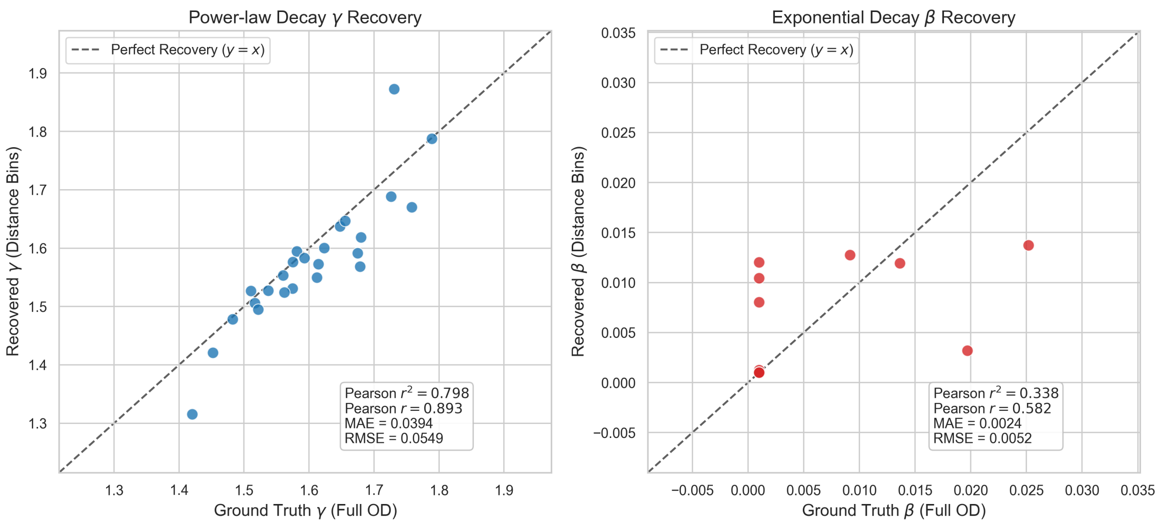

Fig. 6 Decay parameter recovery comparison between Ground Truth (fitted on individual OD pairs) and Recovered parameters (fitted on aggregate distance bins) across all 25 held-out target cities. Power-law exponent γ (left) and exponential decay rate β (right) show consistency in spatial decay structure (Pearson $r = 0.893$ for γ and $r = 0.582$ for β).

It can be seen from Table 8 that the most important is the distance decay $f(d)$: disabling it causes a 31.1% CPC decrease (75.7% of variance explanation). Disabling A_j and O_i results in decreases of 5.3% and 4.7%, respectively (12.9% and 11.4% of variance). The order of importance is always the same for all 25 cities $f(d) \gg A_j > O_i$ and confirms that all three components are important, but the distance decay component is the most important.

Baseline CPC Note: Baseline CPC value of 0.7040 used in this ablation analysis was achieved using fixed national decay prior $(\gamma_{\text{nat}}, \beta_{\text{nat}})$ to freeze the decay component and calculate contributions under a uniform setting. Using the full SDF-ZTF (Survey-Free) framework, where decay parameter values $(\hat{\gamma}, \hat{\beta})$ were estimated separately for each city, CPC performance improves to 0.7124 CPC (as seen in Table 9).

5.4 Evaluating Zero-Shot Transfer Learning Pipeline

We now present an evaluation of the full SDF-ZTF pipeline against the baselines.

5.4.1 Comparison with Baselines

All the models are trained on 25 source cities and tested in zero-shot fashion on 25 target cities. The findings are summarized in Table 9 below.

The Survey-Free version of SDF-ZTF achieves the same performance as DeepGravity (CPC

Table 6 Comparative Outflow Performance (Average CPC across $N = 25$ held-out cities)

Configuration	Mean CPC
Naive Population Baseline	0.584
Predicted Outflow (GBDT)	0.7037
Oracle Outflow (O_i^{true})	0.7447

Table 7 Top-10 Features by Average SHAP Importance (Training Source Cities, $N = 25$)

Rank	Feature	μ_p	S_p
1	total_population	0.232	7.12
2	poi_food_fast	0.084	8.28
3	total_pois	0.057	5.33
4	z_total_pois	0.056	6.45
5	road_density_alt	0.053	2.21
6	road_density	0.050	2.39
7	poi_education_primary	0.030	5.57
8	area_km2	0.025	5.45
9	poi_shopping_supermarket	0.020	8.34
10	poi_entertainment_culture	0.018	3.55

0.7124 vs 0.7117), whereas the Oracle O_i variant of SDF- ZTF achieves CPC 0.7447. Both locally-calibrated versions of Tanner baseline reach CPC 0.750, which acts as a structure-based upper bound, and SDF-ZTF (Oracle O_i) gets close to this upper bound without using any data about target cities.

5.4.2 Statistical Significance Testing

Table 10 presents the results of paired tests for all 50 LOCO folds. DeepGravity has a small but statistically significant advantage under LOCO (ΔCPC $= 0.0255$, $p < 10^{-10}$); however, this advantage evaporates under the 25/25 split.

SDF-ZTF is statistically equivalent to Traditional Gravity ($p > 0.3$) and strongly outperforms the Radiation Model ($p < 10^{-15}$, Cohen's $d = 6.28$).

5.4.3 Urban Morphology Analysis

In the case of overall zero-shot transfer learning per- formance, differences can be observed between spatial performance of the two methods. The analysis for transfer CPC based on urban morphology is shown in Table 11,

where SDF-ZTF works best for coastal and centricities (CPC 0.734 and 0.716), while DeepGravity works marginally better in dense transits (0.695 vs 0.690). It implies that Gravity Decomposition method works well when the distance decay effect dominates.

6 Discussion

6.1 What Aggregate Distributions Reveal About Distance Decay

The results in Section 5 answer the three research questions from Section 1. Aggregate trip-length histograms calibrate the decay function well ($r = 0.893$ for γ; Fig. 6). Distance decay accounts for 75.7% of explainable variance (Table 8). SDF-ZTF achieves CPC of 0.7124 under zero-shot conditions, matching DeepGravity (0.7117) without any target-city OD data (Table 9). In short, a compact set of spatial factors—distance friction, population density, and POI activity—can match the predictive capacity of neural models under cross-city transfer.

6.2 Distance Decay as the Dominant Transferable Mechanism

The 75.7% variance share of $f(d)$ reflects the physical constraint that distance imposes on commuting. While A_j and O_i determine where people want to go and how many trips originate, distance sets the feasible boundaries. Decay rates vary

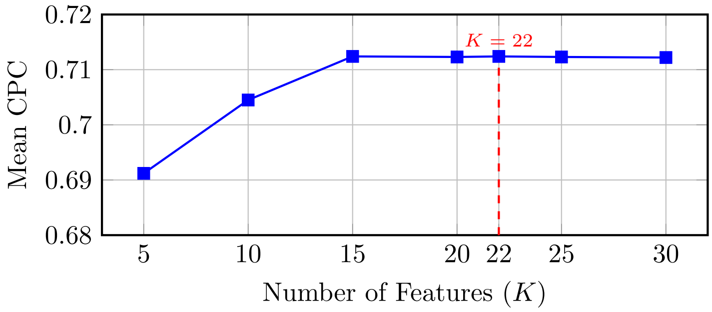

Fig. 7 SHAP feature count performance sweep showing zero-shot with 22 features retains stability

Table 8 Factorial Ablation Results (Average across $N = 25$ held-out target cities)

$\mathbf{O}_i$	f(d)	$\mathbf{A}_j$	CPC	Contribution
✓	✓	✓	0.7040	Baseline (all components)
×	✓	✓	0.671	O_i removal: 4.7% loss
✓	×	✓	0.485	f(d) removal: 31.1% loss
✓	✓	×	0.666	A_j removal: 5.3% loss

across urban layouts (e.g., coastal vs. grid cities), making this the hardest component to transfer. Our results show that aggregate trip-length histograms are sufficient to recover city-specific decay parameters, bypassing the need for individual OD trajectories and enabling privacy-preserving calibration.

6.3 Scientific Implications for Transferable Mobility Modeling

The experimental evidence across 50 US metropolitan areas carries three interconnected implications for how transferable urban mobility should be understood.

On recovering distance decay from aggregate distributions. The findings suggest that individual-level OD trajectories are not necessary to characterize city-specific travel friction. A Pearson correlation of $r = 0.893$ between aggregate-calibrated and OD-fitted γ parameters (Fig. 6) indicates that trip-length histograms—which require only anonymized count data—preserve sufficient information to recover the functional form of distance decay. This implies that privacy-preserving, aggregate data sources can serve as legitimate calibration signals in mobility modeling, with direct consequences for data governance frameworks that restrict access to individual trajectory records. The robustness of this recovery across diverse urban morphologies (Appendix A) reinforces the interpretation that trip-length distributions encode a structurally stable property of urban travel behavior, rather than a city-specific idiosyncrasy.

On the dominance of distance decay among transferable mechanisms. The evidence indicates that distance friction is the primary structural determinant of transferable mobility patterns. With 75.7% of explained variance attributed to $f(d_{ij})$ and a 31.1% CPC drop upon its removal (Table 8), these observations imply that the spatial cost of travel—not destination attractiveness or origin production capacity—is the most universal organizing principle of urban flow. This changes the framing of what makes a mobility model transferable: rather than learning complex interaction effects between urban features, the results demonstrate that faithfully capturing city-specific distance decay accounts for the dominant share of cross-city predictive accuracy. The substantially smaller losses from removing A_j (5.3%) and O_i (4.7%) suggest that these components, while necessary, operate within bounds already constrained by distance structure.

On decomposability and the structural regularity of transferable mobility. Under the zero-shot 25/25 held-out evaluation protocol, the evidence indicates that a multiplicatively

Table 9 Multi-Baseline Comparison (25 Held-Out Target Cities)

Model	Mean CPC	Std Dev	Win % (vs DG)	Type	Features
SDF-ZTF (Survey-Free)	0.7124	0.0329	48.0%	Decomposed	22 (selected)
SDF-ZTF (Oracle O_i)	0.7447	0.028	56.0%	Decomposed	22 (selected)
DeepGravity (E2E)	0.7117	0.0332	–	Neural E2E	30 (all)
Tanner Gravity (Power Const.)	0.750	0.026	72.0%	Structure-only	0
Traditional Tanner Gravity	0.750	0.027	72.0%	Structure-only	0
Traditional Exponential Gravity	0.670	0.035	24.0%	Structure-only	0
Radiation Model	0.512	0.051	0.0%	Param-free	0

Table 10 LOCO 50-Fold Statistical Significance Test Summary

Baseline Model	Mean Diff.	t-stat (t)	p-value (t-test)	Wilcoxon W	p-value (Wilc.)	95% CI Diff.	Cohen's d
DeepGravity	-0.0255	-9.150	3.53×10^{-12}	66.0	3.49×10^{-10}	[-0.0311, -0.0199]	-1.29
Traditional Gravity	0.0000	1.000	3.22×10^{-1}	18.5	3.31×10^{-1}	[-0.0000, 0.0000]	0.14
Radiation Model	0.2492	44.416	3.01×10^{-41}	0.0	1.78×10^{-15}	[0.2380, 0.2605]	6.28

decomposed, interpretable model achieves predictive parity with a neural end-to-end approach: CPC of 0.7124 versus 0.7117 for DeepGravity (Table 9). The results demonstrate that this parity is achieved with only 22 SHAP-selected open-data features, suggesting that the transferable signal embedded in urban form can be accessed without learning distributed latent representations. It is important to note the scope of this claim: under leave-one-out cross-validation (LOCO, 50 folds), a statistically significant gap in favor of the neural baseline persists (ΔCPC $= 0.0255$, $p < 10^{-10}$, Cohen's $d = 1.29$), likely reflecting reduced source-city diversity available per fold rather than a fundamental limitation of decomposed structure. The parity result should therefore be interpreted as specific to the zero-shot transfer regime with balanced training exposure, not as a general performance equivalence claim. Within this scope, these observations imply that transferable mobility patterns may follow a more parsimonious generative structure than has been widely assumed: one in which three separable physical mechanisms—spatial friction, population-driven outflow, and environment-driven attraction—account for the majority of cross-city predictive capacity.

Directions for zero-shot mobility modeling. Taken together, these findings suggest that progress in zero-shot mobility modeling may depend less on architectural complexity than on the identification and faithful representation of stable physical mechanisms. Distance decay, as the dominant transferable component, should be treated not as a nuisance parameter to be absorbed into learned model weights, but as a structurally meaningful quantity that can be recovered from widely available aggregate data. Future work in this area may benefit from investigating the conditions under which the separability assumption holds or fails—particularly in dense, transit-integrated cities where multi-modal interaction effects may violate additive independence—rather than scaling neural capacity to absorb ever-larger feature sets.

6.4 Practical Implications

The decomposed structure offers practical advantages for planning:

Policy Simulation : Planners can modify A_j or O_i independently for "what-if" analyses without confounding the distance decay behavior.
Auditability : Each component is separately auditable, which supports regulatory approval and public accountability.
Survey-Free Deployment : The framework works in data-scarce regions by recovering decay from

Table 11 Zero-Shot Performance by Urban Morphology (25 Held-Out Target Cities)

Morphology Type	# Cities	SDF-ZTF CPC	DeepGravity CPC	Gravity Win %
Dense / Transit	5	0.690	0.695	20%
Sprawling / Grid	11	0.715	0.718	45%
Polycentric	6	0.716	0.712	50%
Coastal	3	0.734	0.712	66%

aggregate histograms and estimating attractions from open data.

6.5 Limitations

1. **Separability Assumption**: The model assumes approximate separability between production, attraction, and decay. In cities with highly integrated multi-modal transit networks or strong spatial interaction effects, this independence assumption may be violated, leading to localized prediction errors.
2. **Geographic Scope**: The evaluation is limited to US metropolitan areas. Cities in other regions (e.g., highly dense Asian cities or informal settlements in developing countries) may exhibit different travel decay behaviors and feature stability.
3. **Temporal Dynamics**: The model uses a pre-pandemic baseline (2019) and does not capture changes in travel friction induced by remote work patterns or modal shifts.
4. **Demographic Representation**: Input datasets like cellular location logs underrepresent elderly, low-income, and smartphone-free populations, introducing potential demographic bias in the output OD flows (Gallotti et al. 2024).
5. **OSM coverage**: Urban areas are comprehensively covered, but the rural fringe has sparse POI data. Transferability to low-density zones remains uncertain.
6. **Causality**: Analysis is purely predictive. Causal effects cannot be inferred (e.g., building a grocery store attracts X more trips). Causal inference requires experimental or quasi-experimental design.

7 Conclusions

This study addressed three questions about transferable mobility structure through zero-shot evaluation across 50 US metropolitan areas.

First, aggregate trip-length histograms are sufficient to calibrate distance decay. The recovered γ correlates well with OD-fitted parameters ($r = 0.893$ across 25 held-out cities), confirming that anonymized distribution summaries serve as an effective calibration signal.

Second, distance decay dominates: it accounts for 75.7% of explainable variance, with removal causing a 31.1% CPC drop—far exceeding the losses from removing attraction (5.3%) or production (4.7%).

Third, the decomposed framework matches neural baselines. Using 22 SHAP-selected open-data features, SDF-ZTF achieves zero-shot CPC of 0.7124, on par with DeepGravity (0.7117) under the same protocol.

These results indicate that transferable mobility structure is more regular than commonly assumed. Identifying stable structural mechanisms from open data offers a transparent and globally deployable alternative to data-intensive neural approaches.

Appendix: Supplementary Materials

Appendix A. Robustness of Aggregate-Based Distance-Decay Recovery

A.1 Estimation Details

Parameters (γ, β) are estimated via MLE from the aggregate trip-length histogram using L-BFGS-B optimization in log-space with 10 random restarts to avoid local optima. Full implementation details are available in the code repository.

A.2 Robustness Analysis

We evaluate the robustness of our MLE parameter recovery under non-ideal data scenarios:

Bin Sensitivity Analysis Sweeping the distance bins $K \in \{3, 5, 10, 20\}$ shows that downstream prediction performance stabilizes at a high level. Specifically, the mean CPC across target cities is 0.7346 for $K = 3$, 0.7367 for $K = 5$, 0.7391 for $K = 10$, and 0.7398 for $K = 20$ (compared to the full OD ground truth baseline of 0.7411). This indicates that even extremely coarse spatial data (e.g., 5 to 10 equal-width bins) is sufficient to robustly capture the decay profile and predict commuting flows.

Missing Attractiveness Data (A_j) Randomly removing 10%, 20%, and 50% of target-city destination attractions A_j yields highly stable decay recovery (e.g., $\gamma_{\text{MAE}} = 0.1312$ under 20% missing data vs. 0.1313 baseline). This validates the robustness of aggregate MLE against localized data gaps.

CBD Collapse (Extreme A_j Distortion) Setting the attraction A_j of the top 5% highest attractiveness zones to zero still allows accurate parameter recovery ($\gamma_{\text{MAE}} = 0.1225$). This proves that spatial decay friction can be recovered even when key destination hubs are omitted.

Appendix B. Robustness of Origin Production Transfer

We analyze the sensitivity of zero-shot production transfer across two dimensions: training-set scale and prediction noise.

B.1 Source-Scale Sensitivity

We analyze how the number of training environments affects zero-shot predictability by retraining the GBDT on subsets of $n_{\text{src}} \in \{5, 10, 15, 20, 25\}$ source cities. Target-city evaluation shows that zero-shot prediction performance rises steeply from $n_{\text{src}} = 5$ and saturates around 15–20 source cities. This indicates that only a modest number of source cities is required to learn stable, generalizable outflow characteristics.

B.2 Noise Sensitivity

To evaluate the stability of origin transfer under prediction errors, we inject additive Gaussian noise at log-scale ($\sigma \in \{10\%, 20\%, 40\%\}$) directly onto the predicted outflows $\log \hat{O}_i$. Even under substantial injected noise (40%), the target CPC degrades only marginally (from 0.7037 to 0.6923). This demonstrates that production-constraint normalization effectively dampens localized estimation noise.

Appendix C. Feature Selection Sensitivity

C.1 Feature Count Sensitivity

Evaluating the GBDT model performance across nested subsets of top SHAP features ($K \in \{5, 10, 15, 20, 22, 25, 30\}$) shows that predictive performance plateaus between 15 and 22 features (CPC: 0.7124). This indicates that most transferable spatial information is captured by a compact set of built-environment indicators, with additional features providing marginal utility.

Appendix D. List of Abbreviations

The key abbreviations and acronyms used throughout this paper are summarized and defined in Table 13.

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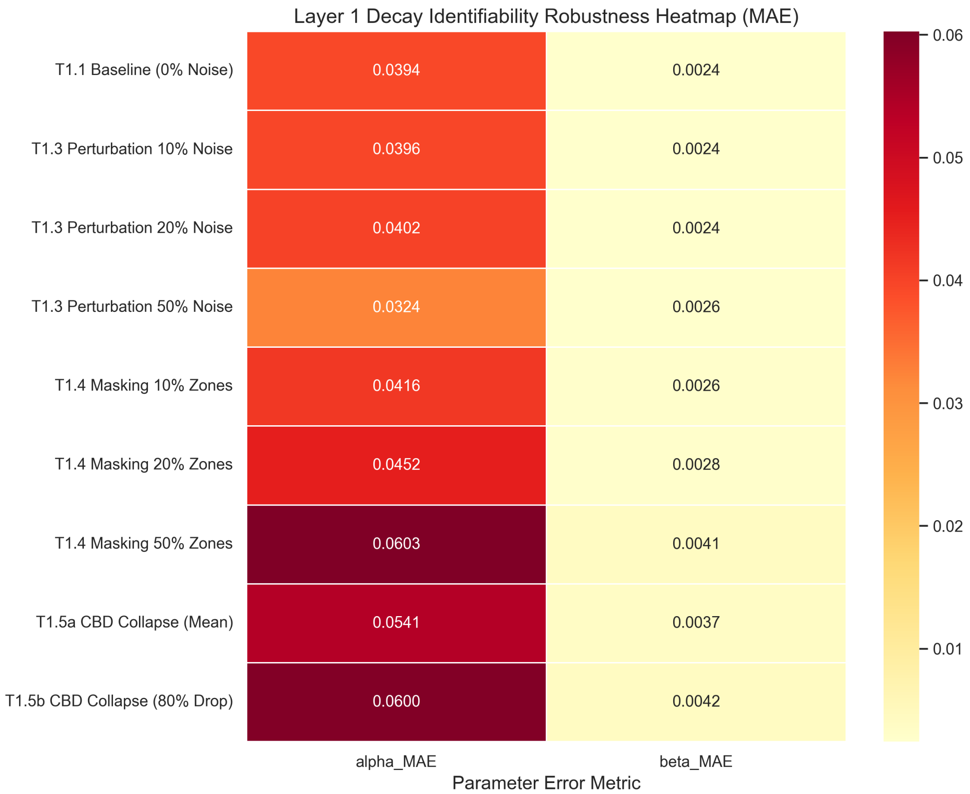

Fig. 8 Robustness of Aggregate-Based Distance-Decay Recovery (MAE) under varying levels of noise, missing data, and CBD collapse scenarios across all 25 held-out target cities.

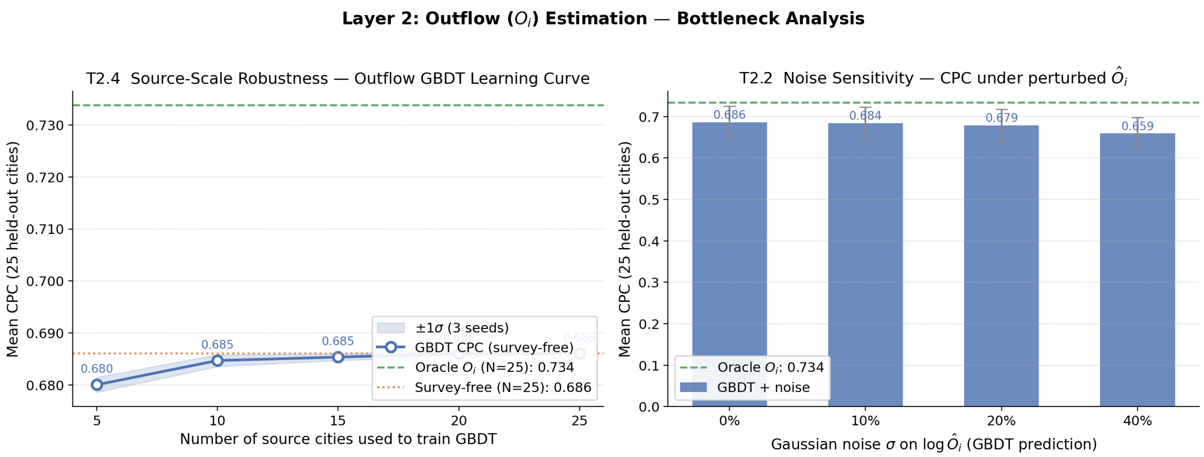

Fig. 9 Outflow bottleneck analysis. **Left**: CPC vs. training source cities (n_{src}), showing performance saturation at 15–20 cities. **Right**: Noise sensitivity of CPC under additive Gaussian noise σ injected on predicted outflows $\log \hat{O}_i$.

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Table 12 GBDT Model Hyperparameter Configurations

Hyperparameter	**Outflow Model (O_i)**	**Attraction Model (A_j, ablation only)**
Base Learner	Decision Tree	Decision Tree
Loss Function	Least Squares (MSE on $\log O_i$)	Least Squares (MSE on A_j)
Learning Rate (η)	0.05	0.05
Number of Estimators	150	200
Max Tree Depth	6	6
Num. Leaves	31	31
Subsample Ratio	0.8	0.8
Colsample By Tree	0.8	0.8
Minimum Child Weight	1	1
Early Stopping Rounds	15	15

Table 13 List of Abbreviations

Abbreviation	**Definition**	**Abbreviation**	**Definition**
CBD	Central Business District	MSE	Mean Squared Error
CPC	Common Part of Commuters	OD	Origin-Destination
GBDT	Gradient Boosted Decision Tree	OSM	OpenStreetMap
GDPR	General Data Protection Regulation	POI	Point of Interest
LOCO	Leave-One-City-Out	SDF-ZTF	Separable Decomposed Framework for Zero-Shot Transferable Flows
MLE	Maximum Likelihood Estimation	SHAP	SHapley Additive exPlanations

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